

Customer Shopping Behavior Analysis

Uncovering insights from transactional data to guide strategic business decisions.



Project Overview & Data Scope



Project Goal

Analyze 3900 purchases to uncover spending patterns and customer segments.



Dataset Size

3900 rows, 18 columns of transactional data.



Key Features

Demographics, Purchase Details, Shopping Behavior, Subscription Status.



Exploratory Data Analysis (EDA) in Python



Data Loading & Exploration

Imported dataset (pandas), checked structure and summary statistics.



Missing Data Handling

Imputed 37 missing 'Review Rating' values using median per product category.



Feature Engineering

Created `age_group` and purchase frequency columns.



Database Integration

Loaded cleaned data into PostgreSQL for SQL analysis.

Key Insights: Revenue & Spending

Revenue by Gender

Total revenue generated by male vs. female customers.

	gender text	revenue numeric
1	Male	157890
2	Female	75191

High-Spending Discount Users

Customers who used discounts but spent above the average purchase amount.

2	64
4	90
9	97
29	94

Target these customers for loyalty programs.

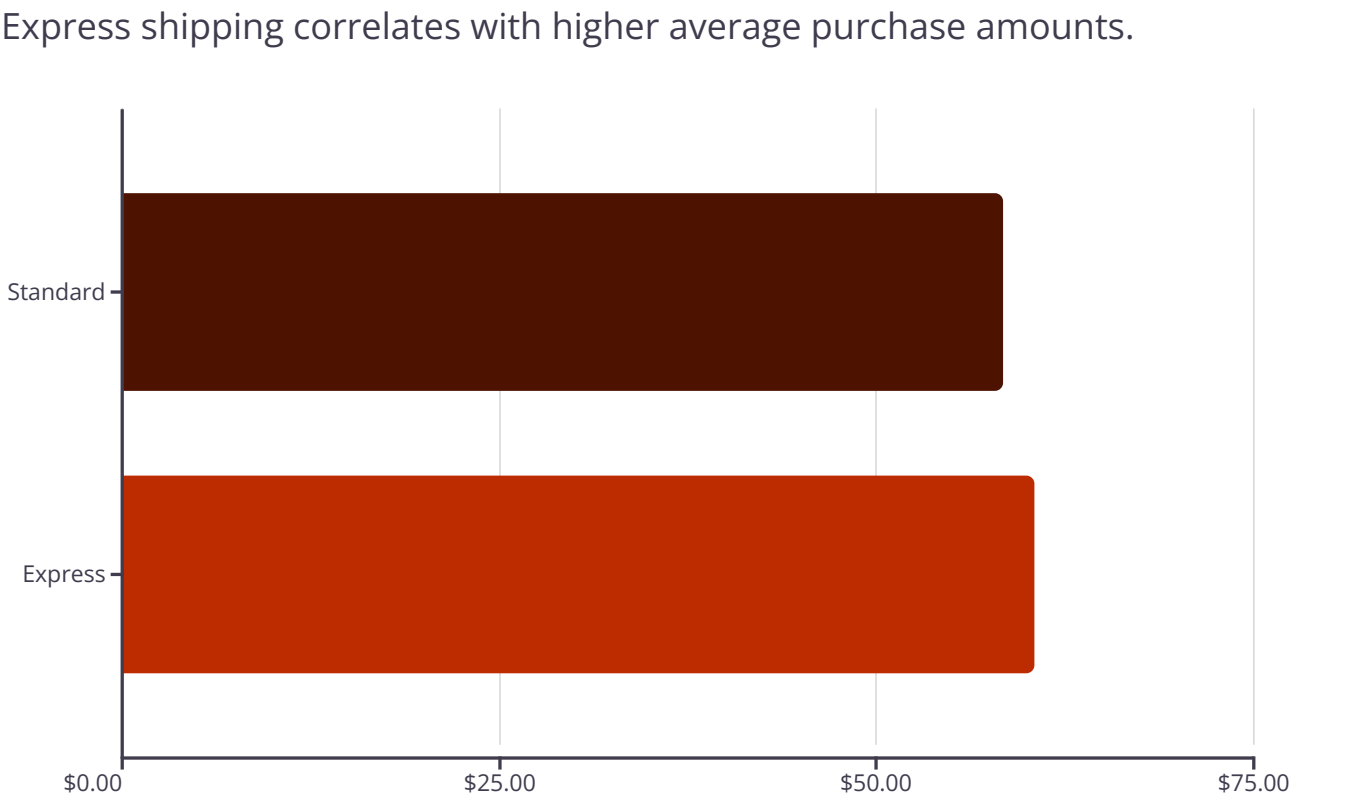
Product Performance & Shipping

Top 5 Products by Rating

Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.79

Highlight these top-rated items in marketing campaigns.

Shipping Type Comparison



Subscription & Discount Dynamics

Subscription Status

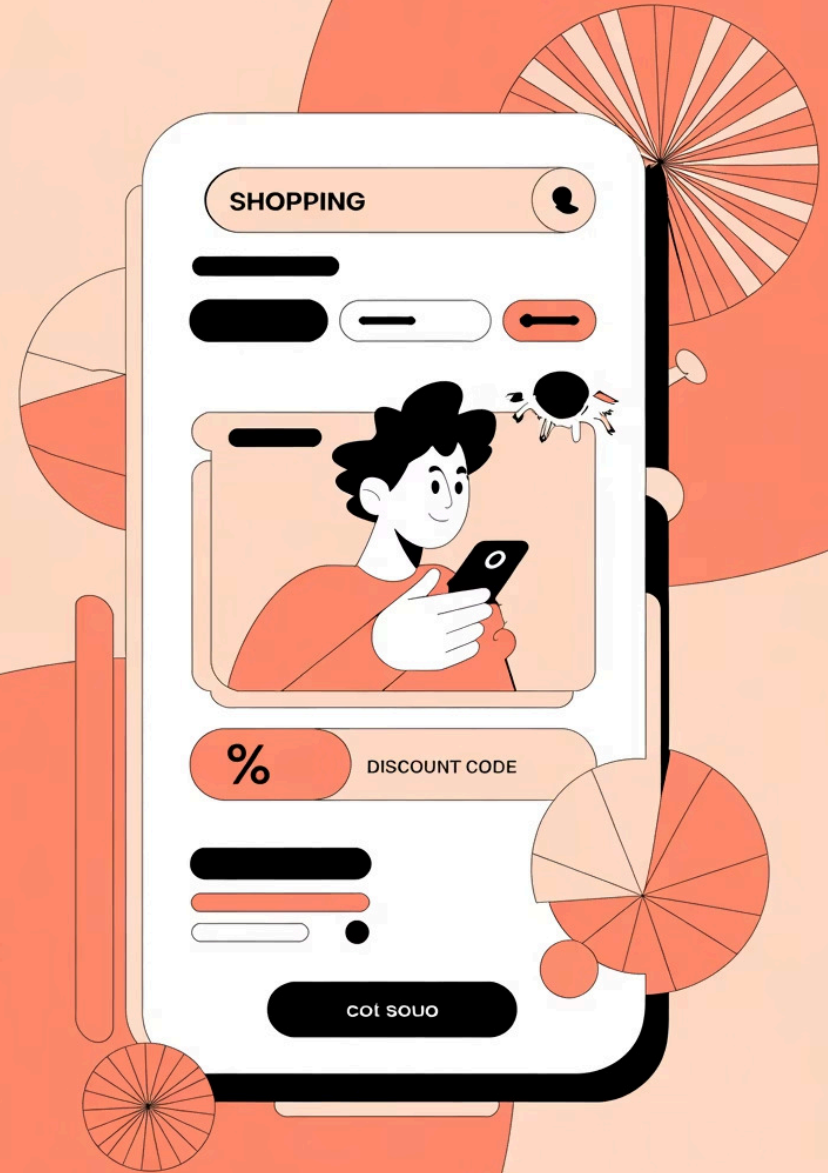
Subscribers (**1053**) have slightly lower average spend (\$59.49) than Non-Subscribers (\$59.87).

Total Revenue

Non-Subscribers drive the majority of revenue: **\$170,436** vs. \$62,645.

Discount-Dependent Products

Hat (50%), Sneakers (49%), and Coat (49%) have the highest percentage of discounted purchases.



Customer Segmentation

Classification based on purchase history reveals a strong base of loyal customers.

3116

Loyal

Customers with significant purchase history.

701

Returning

Customers who have made repeat purchases.

83

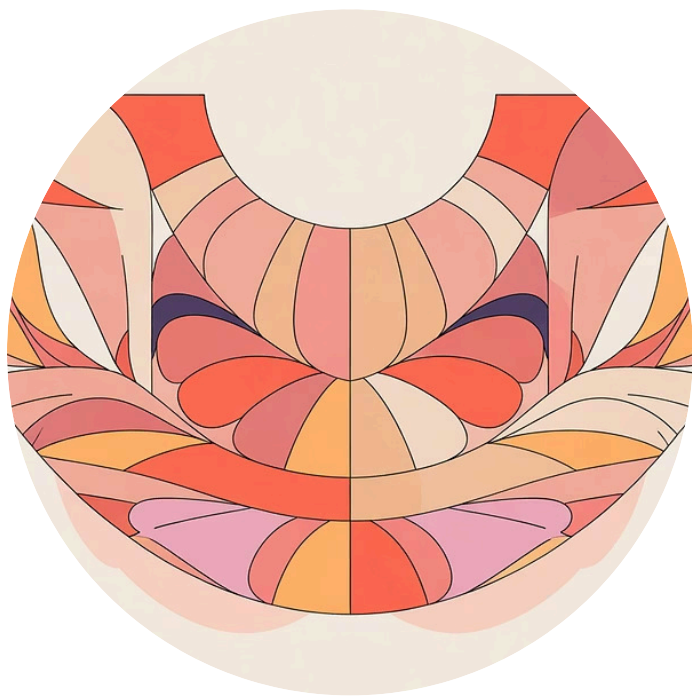
New

First-time buyers in the dataset.

Focus on converting Returning and New customers into the Loyal segment.

Product Preferences by Category

Top 3 most purchased items within Accessories, Clothing, and Footwear.



Accessories

Jewelry (171), Sunglasses (161), Belt (161)



Clothing

Blouse (171), Pants (171), Shirt (169)



Footwear

Sandals (160), Shoes (150)

Age & Loyalty: Revenue Drivers

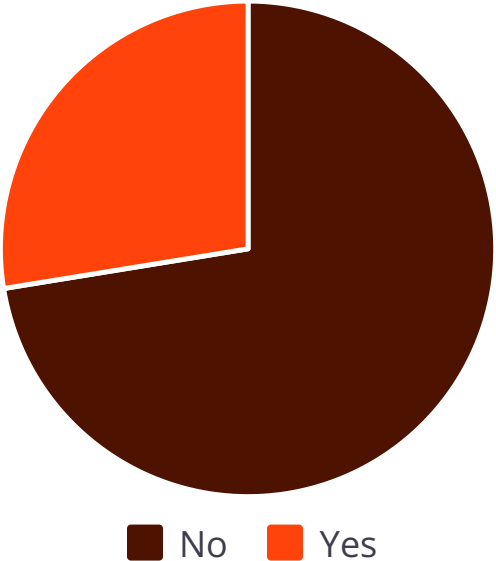
Revenue by Age Group

Young Adults contribute the highest total revenue.

Young Adult	62,143
Middle-aged	59,197
Adult	55,978
Senior	55,763

Repeat Buyers & Subscriptions

Customers with >5 purchases are more likely to be subscribers.



Strategic Business Recommendations



Boost Subscriptions

Promote exclusive benefits to increase subscriber base.



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Customer Loyalty

Reward repeat buyers to solidify the "Loyal" segment.



Review Discount Policy

Optimize discount use to balance sales and margin control.



Product Positioning

Highlight top-rated and best-selling products in all campaigns.

